ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi, Govt. of Karnataka.
Approved by AICTE, New Delhi

Department of Computer Science & Engineering (Data Science)

PROJECT REPORT
ON

From Business Intelligence to Big Data: Concepts, Architecture, Analytics, and Real-World Applications

Subject Name: Big Data Analytics
Subject Code: BDA601

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Submitted To:
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TASK 1: Big Data in Daily Life – Visual Storytelling

TASK 2: BI vs Big Data – Role Play

Characters:
Data Consultant
Business Manager

1. Manager: I don't see why everyone is talking about Big Data. We already use Excel and SQL. They work perfectly.

2. Consultant: They work because your data size is still manageable. But businesses today generate far more data than before.

3. Manager: We handle customer records, sales, and reports. That seems enough.

4. Consultant: That's only structured data. What about customer reviews, social media feedback, images, and website behaviour?

5. Manager: We don't analyse most of that. Is it really important?

6. Consultant: Yes. It shows customer preferences and market trends. Ignoring it means missing opportunities.

7. Manager: But traditional BI gives us monthly and quarterly insights. Why do we need more?

8. Consultant: Because markets change quickly. Companies now need real-time insights to make faster decisions.

9. Manager: Can you give a practical example?

10. Consultant: E-commerce companies track customer activity instantly. They recommend products and adjust pricing in real time.

11. Manager: That sounds useful, but doesn't it increase system complexity?

12. Consultant: It does, but Big Data platforms are designed for scalability and flexibility.

13. Manager: What are the main limitations of traditional BI then?

14. Consultant: It struggles with large data volume, slow processing speed, and limited data formats.

15. Manager: So Big Data solves these issues?

16. Consultant: Exactly. It handles huge datasets, processes them quickly, and supports different data types.

17. Manager: Where does Hadoop come into this?

18. Consultant: Hadoop stores and processes massive data across multiple machines. It reduces cost and improves reliability.

19. Manager: And what about NoSQL databases?

20. Consultant: They store flexible and unstructured data. This is essential for modern applications.

21. Manager: So we don't replace traditional BI?

22. Consultant: No. We combine both. BI handles reporting, while Big Data supports advanced analytics.

23. Manager: What are the real business benefits?

24. Consultant: Better customer understanding, faster decision-making, fraud detection, and improved operational efficiency.

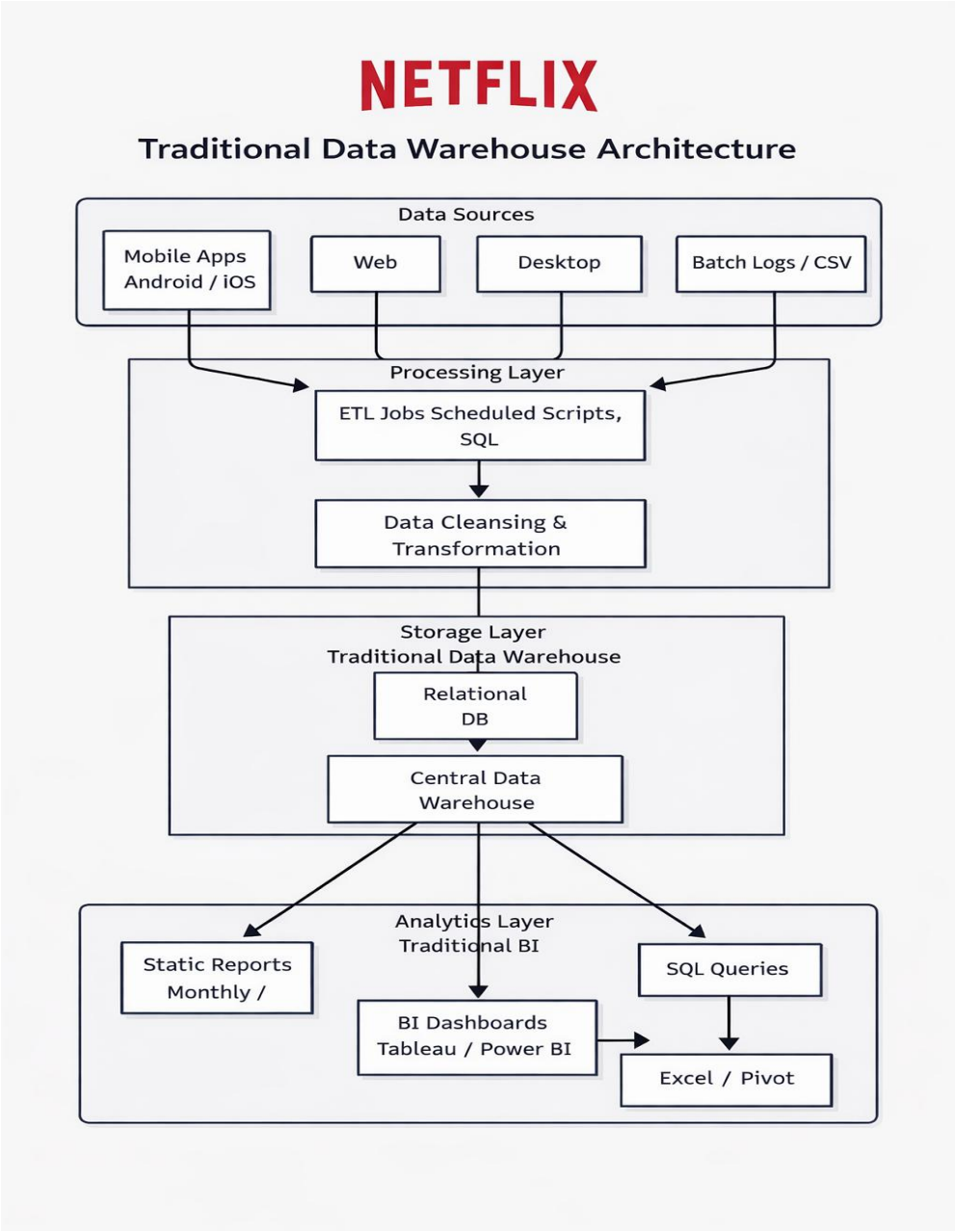
25. Manager: That would help in long-term growth.

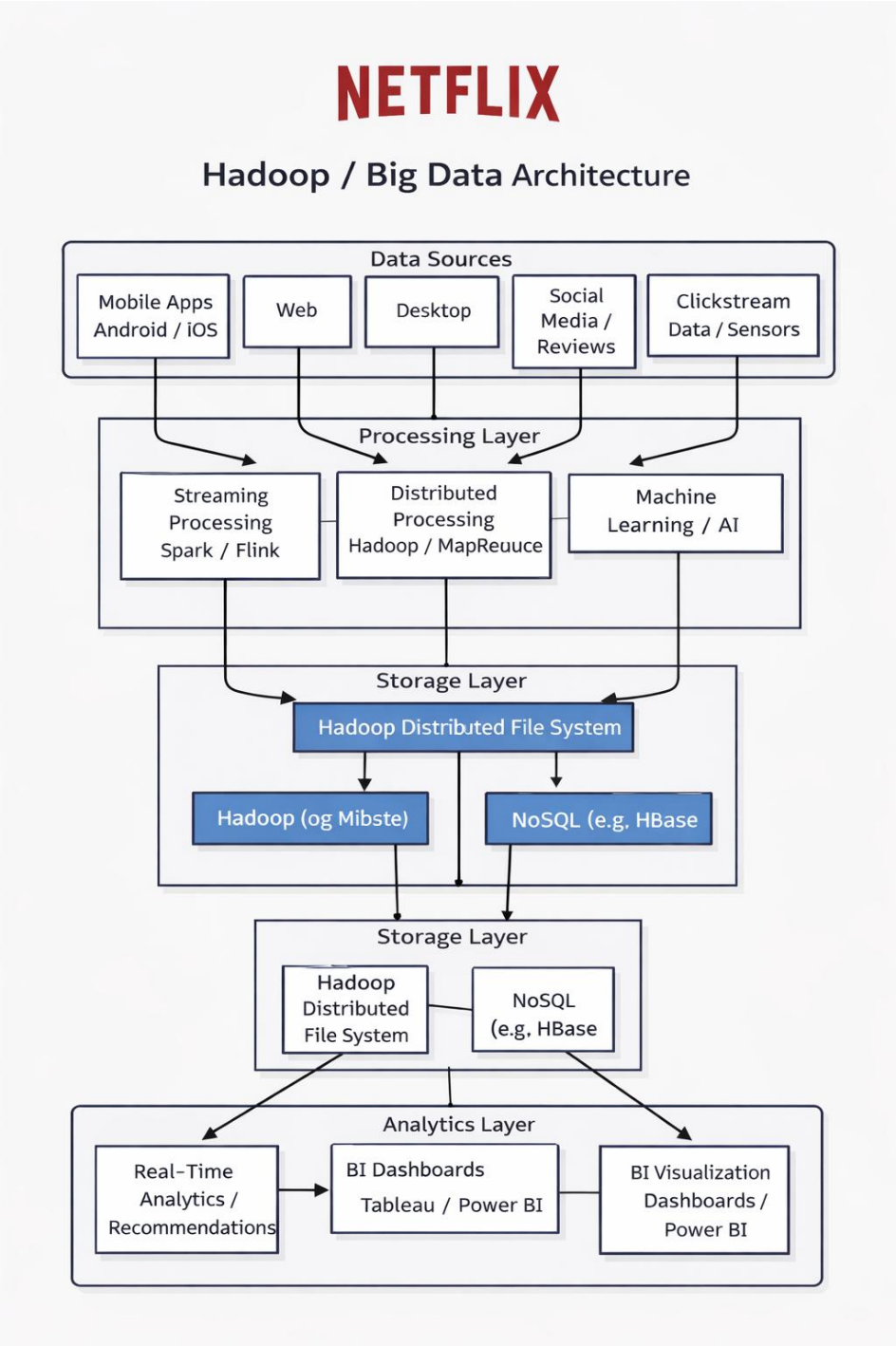
26. Consultant: Exactly. Companies using Big Data gain a strong competitive advantage.

27. Manager: I think we should start exploring it.

28. Consultant: That's the right approach. Early adoption brings better results.

TASK 3: Architecture Design Challenge




NETFLIX
Hadoop / Big Data Architecture
Data Sources
Mobile Apps Android / iOS
Web
Desktop
Social Media / Reviews
Clickstream Data / Sensors
Processing Layer
Streaming Processing Spark / Flink
Distributed Processing Hadoop / MapReuuce
Machine Learning / AI
Storage Layer
Hadoop Distributed File System
Hadoop (og Mibste)
NoSQL (e.g, HBase
Storage Layer
Hadoop Distributed File System
NoSQL (e.g, HBase
Analytics Layer
Real-Time Analytics / Recommendations
BI Dashboards Tableau / Power BI
BI Visualization Dashboards / Power BI

TASK 4: Analytics & Tool Match

Business Question	Analytics Type	Tool
What happened?	Descriptive Analytics	BI Tools (Tableau, Power BI), Hadoop
Why did it happen?	Diagnostic Analytics	Hadoop, NoSQL
What will happen next?	Predictive Analytics	Spark, Hadoop
What action should be taken?	Prescriptive Analytics	Spark, NoSQL, BI Tools

Brief Explanation

1. **Descriptive Analytics (What happened?)**
   This type of analytics focuses on analysing historical data to understand past performance. Business Intelligence tools such as Tableau and Power BI are commonly used to generate reports and dashboards. Hadoop helps in storing and processing large datasets for descriptive insights.
2. **Diagnostic Analytics (Why did it happen?)**
   This involves identifying the root causes of business outcomes by analysing patterns and relationships in data. Hadoop and NoSQL databases support large and complex datasets, enabling deeper analysis of customer behaviour and operational issues.
3. **Predictive Analytics (What will happen next?)**
   Predictive models use historical and real-time data to forecast future trends. Tools like Spark provide faster data processing and support machine learning algorithms to predict demand, customer churn, and risks.
4. **Prescriptive Analytics (What action should be taken?)**
   This type of analytics recommends optimal actions based on predictive insights. Spark, NoSQL, and BI tools help organizations make real-time decisions, improve operations, and enhance customer experience.

Bonus Challenge: "Explain Big Data to a 10-year-old"

Big Data is like a very big basket of fruits. Imagine that every student in your school brings one fruit every day and puts it in the basket. After many days, the basket becomes very full. There are apples, bananas, oranges, and many other fruits mixed together. The basket becomes so big that it is hard to see what is inside or to count each fruit one by one.

Now, if someone wants to know which fruit is the most popular, it will take a long time to check every fruit. So, instead of looking at each fruit slowly, they use a faster and smarter way to sort the fruits. All the apples are kept in one group, bananas in another, and so on. This makes it easy to see which fruit is liked the most.

Big Data works in a similar way. It is about collecting a lot of information and using smart methods to organize and understand it quickly. This helps people make better choices, plan things properly, and solve problems faster. In simple words, Big Data means handling a huge amount of information in a smart and easy way.